

SEMESTER END EXAMINATIONS - JANUARY 2023

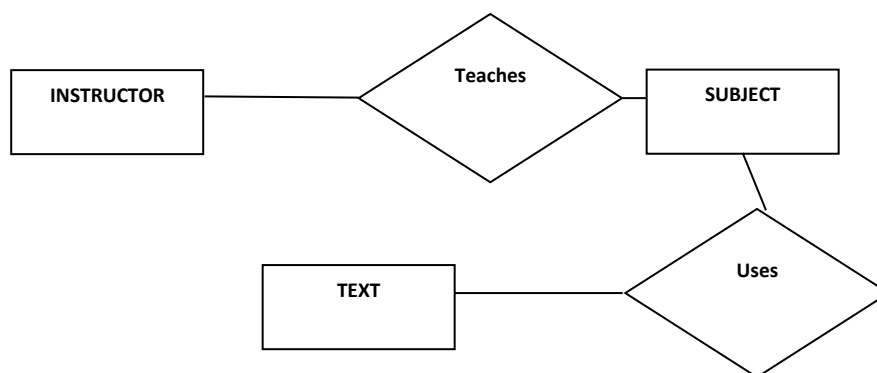
Program	: B.E : Computer Science and Engineering	Semester	: V
Course Name	: Database Systems	Max. Marks	: 100
Course Code	: CS52	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Explain different levels of data abstraction supported by three schema architecture for database systems. CO1 (05)
 - Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Construct the appropriate tables for the E-R diagram obtained, by applying E-R to Relational Mapping rules. Note: Identify minimum three Entity Types. CO1 (10)
 - Differentiate the traditional file system and the database system. CO1 (05)
- Explain any three end users of DBMS with an example for each. CO1 (06)
 - Explain any five advantages of using DBMS approach. CO1 (10)
 - Consider the E-R diagram given below: CO1 (04)

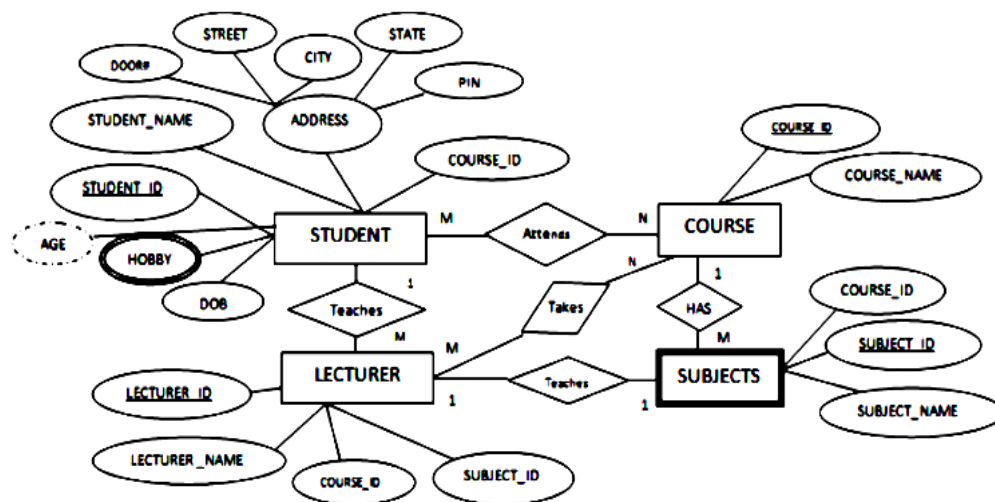


- Supply the attributes for the entity types and mark the key attributes.
- Specify the cardinality ratio and participation constraints in the diagram.

UNIT - II

3.
 - a) Explain the entity integrity and referential integrity constraints with examples. CO2 (06)
 - b) Explain the set operations in Relational Algebra with appropriate examples. CO2 (06)
 - c) Consider the following tables: CO2 (08)
 - EMPLOYEE (person-name, street, city)
 - WORKS (person-name, company-name, salary)
 - COMPANY (company-name, city)
 - MANAGES (person-name, manager-name)
 Write Relational algebraic queries for the following
 - i. Find the names of all employees who work for First Bank Corporation.
 - ii. Find the names and cities of residence of all employees who work for First Bank Corporation.
 - iii. Find the names, street address, and cities of residence of all employees who work for First Bank Corporation and earn more than \$10,000 per annum.
 - iv. Find the names of all employees in this database who live in the same city as the company for which they work.

4.
 - a) Employee(empno, name, office, age) CO2 (06)
 - book(isbn, title, authors, publisher)
 - borrow (empno, isbn, date)
 Write Relational algebraic queries for the following
 - i. Find the names of employees who have borrowed a book published by McGraw-Hill.
 - ii. Find the names of employees who have borrowed all books published by McGraw-Hill.
 - iii. Find the names of employees who have borrowed more than five different books published by McGraw-Hill.
 - b) Explain the mapping of binary 1:N relationship and binary M:N relationship types of ER diagrams to relations. Convert the given ER diagram into relational schema. CO2 (08)



- c) Discuss about the constraints violated during Insert operation? Explain with examples. CO2 (06)

UNIT - III

5. a) Consider the schema for employee database below. CO3 (10)
Employee (employee-name, street, city)
Works (employee-name, company-name, salary)
Company (company-name, city)
Manages (employee-name, manager-name)
Give an expression in SQL for each of the following queries.
i. Find the company that has the most employees.
ii. Find the company that has the smallest payroll.
iii. Define a view consisting of manager-name and the average salary of all employees who work for that manager.
iv. Write CHECK condition to ensure that every employee works for a company located in the same city as the city in which the employee lives.
- b) What is an embedded SQL? For the relational schema given below, write a C function to retrieve the names of pilots certified for a given aircraft using embedded SQL statements. CO3 (10)
Flights(flno: integer, from: string, to: string, distance: integer, departs: time, arrives: time)
Aircraft(aid: integer, aname: string, cruisingrange: integer)
Certified(eid: integer, aid: integer)
Employees(eid: integer, ename: string, salary: integer)
Note that the Employees relation describes pilots and other kinds of employees as well; everypilot is certified for some aircraft (otherwise, he or she would not qualify as a pilot), and onlypilots are certified to fly.
6. a) Consider the following relations containing airline flight information: CO3 (10)
Flights(flno: integer, from: string, to: string, distance: integer, departs: time, arrives: time)
Aircraft(aid: integer, aname: string, cruisingrange: integer)
Certified(eid: integer, aid: integer)
Employees(eid: integer, ename: string, salary: integer)
Note that the Employees relation describes pilots and other kinds of employees as well; everypilot is certified for some aircraft (otherwise, he or she would not qualify as a pilot), and onlypilots are certified to fly.
Write the following queries using SQL:
i. Find the names of aircraft such that all pilots certified to operate them earn more than 80,000.
ii. Print the enames of pilots who can operate planes with cruisingrange greater than 3,000 miles.
iii. Find the eids of pilots certified for Boeing aircraft.
iv. Find the names of pilots certified for Boeing aircraft.
- b) Differentiate nested and co-related nested queries with an example for each. CO3 (04)
- c) Explain the use of group-by and having clause in SQL with an example. CO3 (06)

UNIT- IV

7. a) Discuss the informal design guidelines for relational schema in detail. CO4 (06)
b) What is the need for normalization? Explain 1NF, 2NF, 3NF with examples. CO4 (08)
c) Let $R = ABCDE$, $R_1 = AD$, $R_2 = AB$, $R_3 = BE$, $R_4 = CDE$, and $R_5 = AE$. Let the functional dependencies be:
 $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $DE \rightarrow C$, $CE \rightarrow A$
Apply algorithm to test if the decomposition of R into $\{R_1, \dots, R_5\}$ is a lossless join decomposition or not. CO4 (06)
8. a) A set of FDs for the relation $R \{A, B, C, D\}$ $E: \{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$. Find a minimum cover for this set of FDs illustrating all the inferences used. CO4 (08)
b) Define fully functional dependency. Explain the 2 NF with an example. CO4 (06)
c) Define MVD and explain 4NF related to it with an example. CO4 (06)

UNIT - V

9. a) Explain ARIES recovery algorithm with an example. CO5 (06)
b) What is meant by the concurrent execution of database transactions in a multiuser system? Discuss why concurrency control is needed, and give informal examples. CO5 (08)
c) Write the procedure for recovery based on immediate update in multiple user environment. CO5 (06)
10. a) What is two phase protocol? How does it guarantee serializability? CO5 (06)
b) Why is concurrency control needed? Discuss the types of problems that might be encountered if transactions are allowed to run concurrently. CO5 (08)
c) Explain the recovery techniques based on deferred update. CO5 (06)
